Vatsal Agarwal

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Education

University of Maryland, College Park **Sep. 2022 – May 2028 (Expected)**
Ph.D. Computer Science College Park, MD
Advisor: Abhinav Shrivastava

University of Maryland, College Park **Sep. 2018 – May 2022**
Bachelor of Science in Computer Science College Park, MD

Research Statement

My research focuses on building scalable memory systems for long-context intelligence. I am currently developing memory that lets multimodal models process and reason over video streams from tens of minutes to several hours long. Previously, I worked on multimodal image understanding and panoptic image segmentation.

Professional Experience

TikTok **Sep. 2025 – Mar. 2026**
Research Intern — Matthew Gwilliam *San Jose, CA*

- Developed a training-free memory method for long-form streaming video understanding that improves question-answering accuracy over prior streaming approaches.

Apple **May 2024 – Mar. 2025**
Graduate Machine Learning Intern — Eshan Verma *Cupertino, CA*

- Developed efficient multimodal large-language models leveraging text-to-image diffusion models as question-aware visual encoders, improving performance on vision-centric understanding tasks.

PAII Inc **Jun. 2022 – Nov. 2022**
Graduate Research Intern — Youbao Tang and Mei Han *Remote (San Francisco, CA)*

- Developed a coarse-to-fine Transformer-based method for single-image human mesh recovery, matching prior accuracy with up to 2× fewer parameters and 10× fewer FLOPs.

Selected Publications

- **MMOU: A Massive Multi-Task Omni Understanding and Reasoning Benchmark for Long and Complex Real-World Videos**
  Arushi Goel*, Sreyan Ghosh*, **Vatsal Agarwal***, Nishit Anand*, . . . , Abhinav Shrivastava, . . . , Wei Ping
  NeurIPS 2026 (*equal contribution)
- **Going Down Memory Lane: Scaling Tokens for Video Stream Understanding with Dynamic KV-Cache Memory**
  Vatsal Agarwal, Saksham Suri, Matthew Gwilliam, Pulkit Kumar, Abhinav Shrivastava
  NeurIPS 2026
- **VidParse: Online Parsing of Egocentric Procedures Like a Pro**
  Anubhav Gupta, Archit Kambhamettu, **Vatsal Agarwal**, Pulkit Kumar, Abhinav Shrivastava
  ECCV 2026
- **DiffVLM: Towards Text-Aware Multimodal Understanding via Stable Diffusion**
  Vatsal Agarwal, Matthew Gwilliam, Eshan Verma, Daniel Ulbricht, Abhinav Shrivastava
  WACV 2027
- **MAPS: Memory Augmented Panoptic Segmentation**
  Vatsal Agarwal, Saksham Suri, Max Ehrlich, Abhinav Shrivastava
  ICCV 2025 Workshop: MemVis: The 1st Workshop on Memory and Vision (Oral)
- **Do text-free diffusion models learn discriminative visual representations?**
  Soumik Mukhopadhyay, Matthew Gwilliam, Yosuke Yamaguchi, **Vatsal Agarwal**, Namitha Padmanabhan, Archana Swaminathan, Tianyi Zhou, Abhinav Shrivastava
  ECCV 2024
- **Diffusion Models Beat GANs on Image Classification**
  Soumik Mukhopadhyay, Matthew Gwilliam, **Vatsal Agarwal**, Namitha Padmanabhan, Archana Swaminathan, Srinidhi Hegde, Tianyi Zhou, Abhinav Shrivastava

Awards

CRA Undergraduate Research Award | *Honorable Mention* **Dec. 2021**